Yannic Hartmann

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EDUCATION

Ph.D. in Chemistry (Advisor: Dr. Bernd M. Schmidt) *HHU Düsseldorf*, 2024-present

Thesis: Novel Organic Materials via Dynamic C–C Bond Formation

M.Sc. in Chemistry (Advisor: Prof. Dr. T. J. J. Müller) *HHU Düsseldorf*, 2021-2024

Overall grade: 1.1; Thesis: 1.0 (with distinction)
(German scale 1.0–6.0, where 1.0 = highest)

Ranked 1st in class

Thesis: Synthesis and Photophysical Properties of Ester-substituted Aroyl-S,N-ketene Acetals

Research Stay (Advisor: Prof. Dr. Aiko Fukazawa) *Kyoto University, iCeMS*, 2022-2023

Research Focus: Development of Novel π-conjugated Nitrogen-containing Cross-conjugated Materials

B.Sc. in Chemistry (Advisor: Prof. Dr. T. J. J. Müller) *HHU Düsseldorf*, 2018-2021

Overall grade: 1.2; Thesis: 1.0
(German scale 1.0–6.0, where 1.0 = highest)

Ranked 2nd in class

Thesis: Synthesis and Photophysical Properties of Alkynylated and Aroylated Aroyl-S,N-ketene Acetals

AWARDS AND HONORS

Angewandte Chemie International Edition Poster Award (ISMSC 2026, Bordeaux)	2026
HeRa Travel Grant (ISMSC 2026, Bordeaux)	2026
Kekulé Doctoral Scholarship, German Chemical Industry Fund (FCI)	2025-present
Dr. Manfred Jäger-Prize for the best master thesis	2025
PROMOS Scholarship (High Potential Mobility Grant, DAAD, Kyoto University)	2022
Deutschlandstipendium (Germany Scholarship, Federal Merit Scholarship, annually renewed, partially funded by LANXESS GmbH)	2019-2024

PUBLICATIONS

6) Y. Hartmann, R. Oestreich, Y. Wada, P. de Bary, M. Kawano, C. Janiak, B. M. Schmidt, *Dynamic Covalent Radical Recombination for the Assembly of Tuneable Responsive Porous Organic Cages*, *Angew. Chem. Int. Ed.* **2026**, *65*, e7638565; *Angew. Chem.* **2026**, *138*, e7638565.

5) Y. Hartmann,[+] A. El Abbassi,[+] B. Mayer, U. Resch-Genger, T. J. J. Müller, *Ester-Aroyl-S,N-Ketene Acetals with Solid-State Luminescence: AIEgens from Sequential Three-Component Desymmetrization*, *Chem. Eur. J.* **2025**, *31*, e02071.

4) M. Martínez-Fernández,[+] Y. Hartmann,[+] B. M. Schmidt, *Porous Organic Cages as Building Blocks for Framework Materials*, *Angew. Chem. Int. Ed.* **2025**, *64*, e202509618; *Angew. Chem.* **2025**, *137*, e202509618.

3) J. Voss, Y. Hartmann, E. Nieland, A. Mix, B. M. Schmidt, *Visible-Light Switchable Rings and Chains in Dynamic Covalent Imine Chemistry*, *Chem. Eur. J.* **2025**, e202501047.

2) L. Biesen, Y. Hartmann, T. J. J. Müller, *Alkynylated and triazole-linked aroyl-S,N-ketene acetals: one-pot synthesis of solid-state emissive dyes with aggregation-induced enhanced emission characteristics*, *Sci. Rep.* **2023**, *13*, 14399. DOI: 10.1038/s41598-023-41146-w.

1) L. Biesen, Y. Hartmann, T. J. J. Müller, *Diaroyl-S,N-ketene Acetals: Red-Shifted Solid-State and Aggregation-Induced Emitters from a One-Pot Synthesis*, *Chem. Eur. J.* **2023**, *29*, e202301908.

[+]*These authors contributed equally to this work.*

PROFESSIONAL DEVELOPMENT

Ethics of Leadership Workshop *Wittenberg Center for Global Ethics*, 2025

Intensive 4-day workshop for Kekulé Fellows of the FCI on ethical decision-making, argumentation, and leadership challenges in academic and industry environments

SKILLS AND EXPERTISE

Synthetic Organic Chemistry

Excellent synthetic skills across a broad range of methodologies; Palladium-catalyzed cross-coupling reactions; synthetic planning / retrosynthetic design; Schlenk-technique (inert atmosphere, air- and moisture-sensitive chemistry); Column chromatography; Gel permeation chromatography

Analytical Methods and Material Characterization

UV/Vis spectroscopy; Fluorescence spectroscopy; EPR spectroscopy; NMR spectroscopy; IR spectroscopy; Cyclic voltammetry; Gas sorption analysis

Computational Chemistry

Basic computational chemistry; Geometry optimization; Molecular Modeling

Scientific Illustration and Visualization

Graphic design and 3D rendering for scientific figures (Affinity Designer, Blender)